DAY-1

Dated: 22-07-2021

Introduction to course

Python installation

Python interpreter installation steps

1. python.org>>downloads>>all releases>>windows>>download Pyhton 3.6.8 or 3.7 or 3.8 or any other version as per your requirements(X64 executable file) direct link: https://www.python.org/ftp/python/3.9.0/python-3.9.0rc1-amd64.exe (https://www.python.org/ftp/python/3.9.0/python-3.9.0rc1-amd64.exe)

2. open downloaded file>>RUN>>add python to path(click chechbox)>>install now

   Check whether installaled at correct path

3. This pc>>user>>dell>>app data>>local>>programs>>python>>python 3.x>>scripts>>cmd open.

4. command enter on cmd>>pip install jupyter notebook

5. jupyter notebook (open browser) >>new notebook

```
print('Hello Python')
```

```
Hello Python
```

Day-2

Dated-23-07-2021

```
# print() : is used to display results
# Numbers
print(123)
print(45)
print(676)
# text or symbols or string
# "" or ''
print('Hello python')
```

```python
print("hello Python")
print('This is our 2nd class @ soliatire infosys')
# task: program to print a poem
# program to display your bio-data
```

```
123
45
676
Hello python
hello Python
This is our 2nd class @ soliatire infosys
```

task:

program to print a poem program to display your bio-data

```python
print("POEM\n\nWhere the mind is without fear and the head is held high\nWhere knowledge is free\nWhere the world has not been broken up into fragments\nBy narrow domestic walls\n\nWhere words come out from the depth of truth\nWhere tireless striving stretches its arms towards perfection\nWhere the clear stream of reason has not lost its way\nInto the dreary desert sand of dead habit\n\nWhere the mind is led forward by thee\nInto ever-widening thought and action\nInto that heaven of freedom, my Father, let my country awake.")
```

```
POEM

Where the mind is without fear and the head is held high
Where knowledge is free
Where the world has not been broken up into fragments
By narrow domestic walls

Where words come out from the depth of truth
Where tireless striving stretches its arms towards perfection
Where the clear stream of reason has not lost its way
Into the dreary desert sand of dead habit

Where the mind is led forward by thee
Into ever-widening thought and action
Into that heaven of freedom, my Father, let my country awake.
```

```python
print("INTRODUCTION\n\n\nGood morning sir, First of all thanks for giving me opportunity to introduce my self.\nI am Ritik kashyap.\nI belongs to jalandhar. \nI am currently pursuing Diploma in Computer science and Engineering from Mehr chand polytechnic college, jalandhar.\nI did my schooling from Swami mohan dass model school, jalandhar.\nMy hobbies are playing guitar and playing computers games.")
```

```
INTRODUCTION


Good morning sir, First of all thanks for giving me opportunity to
introduce my self.
I am Ritik kashyap.
I belongs to jalandhar.
I am currently pursuing Diploma in Computer science and Engineering
from Mehr chand polytechnic college, jalandhar.
I did my schooling from Swami mohan dass model school, jalandhar.
My hobbies are playing guitar and playing computers games.
```

Day:3

Date:26/07/21

WORDS OF THE DAY tenable*****to use pescind*****remove parables*****tory vestige*****reminder lamentable*****bad

arithmetic operation

1. addition(+)
2. subtraction(-)
3. multiplication(*)
4. division(/)
5. modules(%)
6. Exponentiation(**)
7. float(//)
8. average

```
# Basic operations

print(12+45+67)
print(34-6)
print(34*2)
print(34/4)
print(34//4)
print(34%4)
print(2**3)

a=45.8
b=67
print(a+b)
print(a-b)

mul=a*b
print(mul)
```

```
print(a*b)
print('product of a and b is: ',mul)
```

```
124
28
68
8.5
8
2
8
112.8
-21.200000000000003
3068.6
3068.6
product of a and b is:  3068.6
```

Task

program to perfrom basic arithmetic operations using varibales

```
print(45+15+78)
print(67*89)
print(78/2)
print(56-24-16)
print(34//43)
print(67%89)
print(6**5)
```

```
138
5963
39.0
16
0
67
7776
```

```
#basic airthmetic operation
a=67
b=45
c=46
d=57
print("addition =",a+b+c+d)

a=36
b=48
c=76
d=67
print("subtraction =",a-b-c-d)
```

```
a=34
b=65
c=42
d=24
print("multipication =",a*b*c*d)

a=22
b=42
c=32
d=34
print("division =",a/b/c/d)

a=32
b=43
c=45
d=56
print("float division =",a//b//c//d)

a=4
b=4
s=a**b
print("exponental =",s)

id(32)
```

Day 4

date:27/07/21

```
# Revision of previous topics
```

task

1. Find the area of rectangle , triangle, square and circle
2. Why python is named as python?

```
# square root

import math
print(math.sqrt(78))
```

```
print(78**0.5)
8.831760866327848
8.831760866327848

#area of circle

R=9
r=6
pi=3.14
print(pi*R*r)


R=9
r=6
areaofcircle=3.14*R*r
print('aoc is:',areaofcircle)
169.56

l=43
b=34
h=43
print("area of rectangle=",l*b*h)

l=94
b=68
print("area of square= ",l*b)

b=78
h=56
print('area of triangle =',h*b/2)

print("The programming language Python was conceived in the late
1980s, and its implementation was started in December 1989[2] by Guido
van Rossum at CWI in the Netherlands as a successor to ABC capable of
exception handling and interfacing with the Amoeba operating system.
[3] Van Rossum is Python's principal author, and his continuing
central role in deciding the direction of Python is reflected in the
title given to him by the Python community, Benevolent Dictator for
Life (BDFL).[4][5] (However, van Rossum stepped down as leader on July
12, 2018.[6]). Python was named after the BBC TV show Monty Python's
Flying Circus.Python 2.0 was released on October 16, 2000, with many
major new features, including a cycle-detecting garbage collector (in
addition to reference counting) for memory management and support for
Unicode. However, the most important change was to the development
process itself, with a shift to a more transparent and community-
backed process.[8]Python 3.0, a major, backwards-incompatible release,
was released on December 3, 2008[9] after a long period of testing.
```

```
Many of its major features have also been backported to the backwards-compatible,while by now unsupported,Python 2.6 and 2.7.")
```

DAY-5

Dated:28-07-2021

```
*Words of the day
1 Quench-----to satisfy thrist by drinking water
2 Reign-----rule
3 Sovereign---ruler
4 Descendent----successor
5Plunge-----to jump or drive quickly and energetically
```

```
  File "<ipython-input-25-9195eb8e0b27>", line 1
    *Words of the day
           ^
SyntaxError: invalid syntax
```

```
# Variables: are the containers to store data
# dynamically typed language
# syntax: variable_name = data
a=243
b='hello python'
c='d'
d='$%^&&*'
# display results
print(a)
print(b,c,d)
# id(): is used to find the memory address of data
print(id(a))
print(id(c))
# Python is an interpreted language
```

```
id('hello')
```

```
a='hello'
id(a)
```

```
a=35
b=67
# addition
add=a+b
print(add)
# subtarction
print(a-b)
# Mulplication
```

```
print('The Product of a and b is:',a*b)
print(a/b) # division
print(a//b) # floor division
print(a%b) # modulus
print(a**b) # exponent
print(2**4)

# avegrage : (sum of all obssevations)/total no. observations
print((23+45+7+56)/4)

# PEMDAS: Parenethesis, exponent ,muplication, division, addition , subtract
```

Task

1. What is zen of python?

```
The Zen of Python is a collection of 19 "guiding principles" for
writing computer programs that influence the design of the Python
programming language.[1] Software engineer Tim Peters wrote this set
of principles and posted it on the Python mailing list in 1999.[2]
Peters's list left open a 20th principle "for Guido to fill in",
referring to Guido van Rossum, the original author of the Python
language. The vacancy for a 20th principle has not been filled.

Peters's Zen of Python was included as entry number 20 in the
language's official Python Enhancement Proposals, which was released
into the public domain.[3] It is also included as an Easter egg in the
Python interpreter, which can be displayed by entering import this

  File "<ipython-input-4-9fb5f1029a9c>", line 1
    The Zen of Python is a collection of 19 "guiding principles" for
writing computer programs that influence the design of the Python
programming language.[1] Software engineer Tim Peters wrote this set
of principles and posted it on the Python mailing list in 1999.[2]
Peters's list left open a 20th principle "for Guido to fill in",
referring to Guido van Rossum, the original author of the Python
language. The vacancy for a 20th principle has not been filled.
        ^
SyntaxError: invalid syntax


2. what is pep-20?

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
```

```
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!

3. Take atleast four variables and perfrom basic arithmetics
operations on them

a=67
b=45
c=46
d=57
print("addition =",a+b+c+d)

a=36
b=48
c=76
d=67
print("subtraction =",a-b-c-d)

a=34
b=65
c=42
d=24
print("multipication =",a*b*c*d)

a=22
b=42
c=32
d=34
print("division =",a/b/c/d)
```

DAY- 6

DATED:29-07-2021

```
# ZEN OF PYTHON
import this
```

```
# integers
i=35
print(i)
# type(): used to check the class of data
print(type(i))
# id()
print(id(i))

# floting point (float)
f=45.5757456474654666666666666666666666666666666666666666
print(f)
print(type(f))
print(id(f))
# type-casting: conversion of one type of data into another type
i=float(i)
print(i)
print(type(i))
# Multiline text/comments
print('''Python is an
 interpreted language''')
print("""Python is an
 interpreted language""")
```

Task

1. program to print a poem using multiline commenting statement

```
#Program to print a poem using multiline commenting statments

print("""Where the mind is without fear and the head is held high
Where knowledge is free
Where the world has not been broken up into fragments
By narrow domestic walls

Where words come out from the depth of truth
Where tireless striving stretches its arms towards perfection
Where the clear stream of reason has not lost its way
Into the dreary desert sand of dead habit

Where the mind is led forward by thee
Into ever-widening thought and action
Into that heaven of freedom, my Father, let my country awake.""")

#2. program to print bio data of your's using multiline commenting
statement

#program to print bio data of your's multiline commenting statements

print("""INTRODUCTION
```

```
Good morning sir, First of all thanks for giving me opportunity to
introduce my self.
I am Ritik kashyap.
I belongs to jalandhar.
I am currently pursuing Diploma in Computer science and Engineering
from Mehr chand polytechnic college, jalandhar.
I did my schooling from Swami mohan dass model school, jalandhar.
My hobbies are playing guitar and playing computers games.""")

#3. program to add int and float and display results using string
concatenation

#program to add int and float and display them result using string
concatencation

a=8
b=6.64
c=4
d=9.38
print('add abcd:',a+b+c+d)
```

Day-7

Dated: 30-07-2021

```
#words of the day
1. confide---trust
2. custom---habit
3. converge---combine
4. catalyst---something that precipitates an event
5. catastrophe---calamity

# Solution
# program to add int and float and display results using string
concatenatio
a=34
b=45.568
add=a+b
print('The sum of a & b is:',add)
print('The sum of a & b is:',a+b)

# input(): used to take input data from the user
a=int(input('Enter any value here: '))
print(a)
b=float(input('Enter any value here: '))
```

```
print(b)
print(a**b)

Enter any value here:

------------------------------------------------------------------------
-----
ValueError                                Traceback (most recent call
last)
<ipython-input-26-8bd4b2b34696> in <module>
      1 # input(): used to take input data from the user
----> 2 a=int(input('Enter any value here: '))
      3 print(a)
      4 b=float(input('Enter any value here: '))
      5 print(b)

ValueError: invalid literal for int() with base 10: ''

list=['a','b','c','d']
list=[x for x in list if ord(x)>97]
print(list)
```

Task

1. make an Arithmetic calculator and take numbers as user input (use string concatenation)

```
# MAKE A ARITHMETIC CALCULATOR AND TAKE NUMBER AS USER INPUT(USE
STRING CONCATENTATION)

a=int(input('enter the value of a')) # use int for calculation
b=int(input('enter the value of b'))
add=a+b
print('addition of a and b:',a+b)
a=int(input('enter the value of a '))
b=int(input('enter the value of b'))
sub=a-b
print('subtraction of a and b:',a-b)
a=int(input('enter the value of a'))
b=int(input('enter the value of b'))
mul=a*b
print('the multiplication of a and b :',mul)
a=int(input('enter the value of a'))
b=int(input('enter the value of b'))
div=a/b
print('the divison of a and b :',div)
```

DAY-8

DATED:02-08-2021

```
*****WORDS OF THE DAY*******
Cadet----recruit
Eavesdrop---listen in
Manipulative---infuence
Malfunction----glitch
Misfortune---mishap

# Mulitple user input

**syntax:
 v1,v2,v3...,vn= input('Enter the values: ').split('seperator')

# emoji print()
Syntax:
 "\unicode"
 OR
 "\N(name of emoji)"

#Prateek Kumar9:06 AM
#task
a,b,c,d,e=input("enter any five values:" ).split()
print(a,b,c,d,e)
a=int(a)
b=int(b)
c=int(c)
d=int(d)
e=int(e)
print("the sum of a,b&c is:",a+b+c+d+e)
print("the multiplication of b,c,d is:",b*c*d)
```

TASK

1. Make an Arithmetic calculator and take FIVE numbers as MULTIPLE user input (use string concatenation with emoji's)
2. print() your data like name, age, technology using multiline string and also use emoji's
3. What is antigravity module?

```
# make a arithmetic calculator and take five numbers as MULTIPLE user input(use string )

a,b,c,d,e= input("enter the value:").split("+")
print('addition :',int(a)+int(b)+int(c)+int(d)+int(e))
a,b,c,d,e=input('enter the value').split("*")
mul=int(a)*int(b)*int(c)*int(d)*int(e)
```

```
print('multiplication solution :',mul)
a,b,c,d,e=input('enter the value:').split('-')
sub=int(a)-int(b)-int(c)-int(d)-int(e)
print('solution:',sub)
```

```
#print()your data like name,age,technology using multiline string and
also use emoji's
print(''' my name is RITIK KASHYAP,'\U0001F497'
 i m 18 yrs old,'\U0001F92B'
 i m persuing in mehr chand polytechnic college,('\U0001F393')''')
```

```
#what is antigravity module?
```

```
 The antigravity module, referencing the XKCD comic mentioning Python,
was added
to Python 3 by Skip Montanaro. Update: The Python 3 stdlib version has
an easter egg
inside the easter egg, if you inspect the source code: it defines a
function that
purports to implement the XKCD geohashing algorithm.
```

DAY 9

DATED:02-08-2021

```
      Word of the day
hover***hang
humiliate***shame
igneous***fiery
innovation***new
limitation***copy
```

```
### Strings:
1.Strings are the combination of characters
2.Strings are always defined inside single or double codes('',"")
3 Strings are ordered collection of elements
4Syntax
 variable='String' or"String" 'string'
```

```
# Definition
print('hello')
print("hello")
# with variable
st=('Hello python')
st_1=("hello python")
print(st)
print(st_1)
```

```
str='hello'
str[:1]

st='hey hello python'
print(st[2])
print(st[-14])

print(st[4:9:1])
print(st[::])

print(st[-1:-10:-1])
print(st[::-1]) #reverse string

#len(): used to find the number of elements of sequence
print(len(st))

print(len('hello'))
```

task

1. take a string as user input and calculate its length
2. typecast your string into numbers like int,float
3. calculate reverse of user defined string

```
#TAKE A STRING AS USER INPUT AND CALCULATE ITS LENGTH

st=input('enter the value:')
print(len(st))

#TYPE CAST YOUR STRING INTO NUMBERS LIKE INT,FLOAT

a=input('enter the value')
a=float(a)
print(a)
a=int(a)
print(a)

#CALCULATE THE REVERSE OF USER DEFINED STRING

st='RITIK KASHYAP'
print(st[3:10:1])
```

Day 10

word of the day

jolt = shock junction = intersection jeoparadize = threaten jab = dig anxious = jealousy

Function of string

```
# len ()
string='morn@ing batch python'
print(len(string))
print(string)

# capitalize ()
print(string.capitalize())

# title()
print(string.title())

# upper () and lower ()
print(string.upper())
print(string.lower())

print(string.islower())
print(string.isupper())

# index () and find ()
print(string.index('b'))

print(string.index('a'))


print(string.find('l'))

# count ()
print(string.count('o'))

# enumerate
print(list(enumerate(string)))
print(enumerate(string))

# replace ()
print(string)
string.replace('@','')

# center ()

string.center(len(string)+6,'\U0001F600')

user=input('enter no. of character to print:')
string.center(len(string)+6,'\U0001F600')

#Task
1. perform all remaining methods of string
2. take a string as user input and replace a user defined character
with user defined character
3. take a user input and print its len, and reverse only if all the
character are alphabets
```

```
4. make a python program to print user defined characters around the
string with user defined length

input_string = str(input("ENTER  A STRING :"))
character = input("enter a character you want to modify in above
string:")
symbol = input("Enter the symbol you want to replace with :")
modified_str=input_string.replace(character,symbol)
print("modified string is :",modified_str)

dir(str)

#capatalise
a="bee string"
print(a.capitalize())

#casefold
a="BEE"
print(a.casefold())

#center
a="bee"
b=a.center(12,"-")
print(b)

#QUES: PERFORM ALL REMAINING METHOD OF STRING
str='my name'
print(str.isalnum())
print(str.isalpha())
print(str.isascii())
print(str.isdecimal())
print(str.isdigit())
print(str.isidentifier())
print(str.isnumeric())
print(str.isprintable())
print(str.isspace())
print(str.istitle())
print(str.isupper())

print(str.format())

#QUES: TAKE A STRING AS USER INPUT AND REPLACE USER DEFINED CHARACTER
WITH USER DEFINED CHARACTER
str=input('enter the value:')
print(str)
print(str.replace('l',''))

#QUES: TAKE A USER INPUT AND PRINT ITS LEN,AND REVERSE ONLY IF ALL THE
CHARACTER ARE ALPHABETS
str=input('enter the character')
```

```
print(len(str))
print(str[::-1])

#QUES: MAKE A PYTHON PROGRAM TO PRINT USER DEFINED CHARACTER AROUND
THE STRING EITHER USER DEFINED LENGTH
a=input('enter the value')
print(a.center(len(a)+6,'\U0001F603'))
```

Day 11

```
**word of the day**
habitat   = surrounding
hack      = unauthorise
homicide  = murder
hail      = cheer
knuckle   = part of the hand

List data type
list is an ordered sequence of elements
list can contain different forms of data like int, float, string, char
syntax:
   variable=[e1,e2,e3,.....,en]

#definition
list_1=[1,2,4.5,'hgj','e','a']
print(list_1)

print(type(list_1))

#empty list
empty_list=[]
print(empty_list)

#id ()
print(id(list_1))
print(id(empty_list))

len ()
print(len(list_1))
print(len(empty_list))

print(list_1)

list_1[2]


list_1[-3]

#take a list as input from user and calculate reverse of list
```

```python
user_list=input('enter the elements of list:').split()
print(user_list)

print(user_list[-1:-1])
print(user_list[::-1])

#solution
user_list=input('enter the elements of list:').split()

#solution
user_list=input('enter the elements of list:').split()

dir(list)

#adding elements to list

list_2=[]
print(list_2)
list_2.append(123)
list_2.append('gh')
list_2.append('python')
list_2.append(12.3)
print(list_2)

#extend ()

list_2.extend('string')
list_2.extend([1,2,3,4])
print(list_2)

#update

list_2[1]='java'
print(list_2)

#remove elements from list

print(list_2)
list_2.pop()
list_2.pop(1)
print(list_2)
```

Day 12

```
   ***word of the kinethetic***
kinesthetic = physical
kernal      = core
laaison     = communication and cooperation
lukewarm    = warm
lethal      = something that can cause death
```

```
 list_3=[123, 'gh' ,'python', 12.2, 's', 't', 'r', 'i', 'n', 'g',]
print(list_3)

#remove()
list_3.remove('gh')
list_3.remove(12.2)
print(list_3)

#clear()
list_3.clear()
print(list_3)

#del()
del(list_3)
print(list_3)

#reverse()
list_4=[123, 'python', 's', 't', 'r', 'i', 'n','g', 1,2,3,4,5,6,7,8,9]
print('original list is:' , list_4)
list_4.reverse()
print('reversed list is: ', list_4)

#sort()
list_4.sort()
print(list_4)

list_5=[7,8,9,5,5,4,3]
list_5.sort()
print(list_5)

list_6=['s','t','d','s','s']
list_6.sort()
print(list_6)
list_6.reverse()
print(list_6)

#task in class
print a sorted list using upper and lower case letter

list_7=['R','Y','y','a','U']
list_7.sort()
print(list_7)

#sublists
list_8=[1,2,[4,5,6],[6,3,5,[6,9,0]],2,5,[7,8,9]]
print(list_8)

#task
take a list of character consisting upper and lower case and start it

##tuple
```

```
1. tuple is an oredered sequence of element
2. tuple is imutable data type
syntax:
        variable = (e1,e2,e3,.....en)
        variable =  e1,e2,e3,.....en
```

```
#
tuple_1=(1,2,3,4)
print(tuple_1)
tuple_2=(5,6,7,8,9,0)
print(tuple_2)

print(type(tuple_1))
print(type(tuple_2))
```

task

1. take tuple as user input and perfome basic functions on it
2. calculate the reverse of tuple
3. define a tuple with one element

```
# TASK 1.TAKE A TUPLE AS USER INPUT AND PERFORM BASIC FUNCTIONS ON IT
tuple=input('enter the value :')
print(type(tuple))
print(id(tuple))
tuple_1=int(input('enter the value:'))
tuple_1=float(tuple)
print(tuple_1)

#2.CALCULATE THE REVERSE OF TUPLE.
tuple=45,485,5849,47964,4885
print(tuple[::-1])

#3.DEFINE TUPLE WITH ONE ELEMENT.
tuple=(10)
print(tuple)
```

DAY 13

```
 **word of the day**
1. Monument = Masterpiece
2. Madeleine = cookie
3. Fumble = Goof uprenetic
4. Frenetic =Excited
5. Flinched = Cringe
```

```
dir(tuple)

tuple_3=(1,2,3,4,5,6,1,2,3,2)
tuple_3.count(1)

# task
t=(1,2,3,4)
# to add (2,3,4)

t=(1,2,3,4)
T=(2,3,4)
tuple=t+T
print(tuple)

t=(1,2,3,4)
t1=(2,3,4)
# typecasting
l=list(t)
l.extend(t1)

t=tuple(l)
print(t)

t=t1
```

Control Statement

1. if statement: Syntax: if condition: body
2. if else: Syntax: if condition: body else: body
3. elif statement Syntax: if condition1: body elif condition2: body elif condition3: body

```
a=12
b=12
if a==b:
    print('Yes')

age=int(input('Enter your age'))
if age>15:
    print('Elder')
else:
    print('Child')

age=int(input('Enter your age:'))
if age<15:
    print('child')
elif age>=15 and age<30:
    print('Youngster')
```

```
else:
    print('Child')

Task
Make a program to mark a grade to a student according to his
marks(enter marks as input).

marks=int(input('Enter the marks:'))
if marks<20:
    print('Fail')
if marks<30:
    print('C')
if marks>=40:
    print('B')
if marks>45:
    print('A')
else:
    print('Fail')

# Swapcase
a=12
b=45
print('a:',a)
print('b:',b)
a,b=b,a
print(a)
print(b)
```

task

1. make a verification system to allow user to vote
2. make vending machine
3. make a program to create a simple chat bot

```
#1.Make a program to grade to a student according to his marks (take
number as input)
RITIK=int(input('marks obtained:'))
sid=int(input('marks obtained:'))
if RITIK>sid:
    print('first')
else :
    print('second')

# swap 2numbers

a=int(input('enter the number:'))
b=int(input('enter the number:'))

a,b=b,a
```

```
print(a)
print(b)

# task

#1.make a verification system to allow user to vote
person=int(input('enter your age:'))
if person>18:
    print("you can vote")
else:
        print("you can't vote")

#2.make a vending machine
print('welcome to the system')
items=int(input('enter the amount:'))
if items==5:
    print('kurkure')
elif items==20:
    print("lays")
else:
    print('try again')

#3.make a program to create a simple chat bot
user=input('say any thing:')
if user=='hello' or user=='hey' or user=='gud mrng':
    print('hey there,how are u')
elif user=='name' or user=='who are u':
    print('i m ur assistant,how can i help')
elif user==exit or user==by:
    print("gud day sir")
else:
    print('invalid input')
```

DAY 14

```
**Word of the day **
Capricious = Changeable
Attenuate = Weakened
perambulate = proceed
Suspicious = Doubtful
Parry =  Repel

###loop statement
1. while loop
          syntax:
              while condnition:
                body
                increment/decrement
```

```
2. for loop
          syntax:
            for iterator in sequence:
        body


#task

a=2
while a<31:
    print(a)
    a=a+2

# infinite loop
while True:
    print('hello')

# for loop
for s in 'string':
    print(s)

for i in [1,2,3, 3 ,5,6,7,8]:
    if i%2==0:
    print('even')
    else:
    print('odd')
```

Task

```
1. program to take a list from user as input and find out even and odd
numbers from it
2. Find out the vowels and count total no.of vowels occuring in a
string
3. program to find whether a string or a number is palendrome or not
ram --mar, kanak --kanak

#program to take a list from user as input and find out even and odd
numbers
list=int(input('enter the number:'))
mod=list%2
if mod>0:
    print('odd')
else:
    print('even')

#find out the vowels and count total no.vowels occuring in a string
 sentence=input("entr the sentence:")
string=sentence. lower()
print(string)
```

```
count=0
list1=["a","e","i","0","u"]
for char in string:
     if char in list1:
        count=count+1

print("number vowels in given sentence is:",count)
```

```
#program to find whether a string or a number is palendrome or not
a=input('enter string:')
b=a[-1::-1]
if(a==b):
    print('palindrome string:')
else:
    print('not palindrome string:')
```

DAY 15

```
** WORD OF THE **

Neglect = Not to pay attention
Native = Birh place
Numb = To loose sensation of body
Nightmare = A bad dream
Optimum = Best
```

```
#solution
#program to take a list from user as input and find out even and odd
numbers from it
user_list=input('enter element of list :').split()
print(user_list)

for i in user_list:
    if int (i)%2==0:
        l1.append(i)
    else:
        l2.append(i)
print('list of even number is:',l1)
print('list of odd number is:',l2)
```

```
#program to find whether a string or a number is palendrome or not ram
--mar, kanak --kanakput()
user_string=input('enter a string: ')
reverse=user_string[::=1]
if user_string==revese:
     print('the given string is a palendrome.')
else:
    print('the given string is not palendrome,')
```

```
#Range()
#syntax: range(start value, end value)
var=range(1,30)
for i in var:
    l1.append(i)
print(l1)
```

```
#Random module
import random
num=random.randrange(1,6)
print(num)
```

```
var=random.choice([1,2,3,4,5,6])
print(var)
```

```
import calendar
cal=calendar.calendar(2021)
```

task

```
#1=program to simulate a dice
```

Day 16

```
**word of the day**
obsruct = blocking
obnoxious = unpleasent
overwhelm = submerged
omission = left out
petrify = paralyzed
```

TEAM INFORMATION

```

```

Day 17

```
**word of the day**
quest = to search something
conquest = victory
dribble = soar
crumble = to break
stratle = astonish
```

FIRST TEST

Day 18

```
**word of the day
Resilient = flexible
Residual = remains
Squad = band
substitute = replace
Sophomore = third year student

def add(a,b):
    return  'The sum of a nd b is:',a+b

add(2,75)

add(45,78)

add(57.45,12.78)

# task
Make a function to find even numbers

def even_odd(a):
        if a%2==0:
            return f' {a} is even no'
        else:
             return f'{a} is odd no'

def even_odd(a):
        if a%2==0:
            return f' {a} is even no'
        else:
             return f'{a} is odd no'

print(even_odd(17))

# f concatentation
# syntax
name='book'
var=4
na=77
print(f'this is a {name} and {var} b {na}')
```

Day 19

```
**word of day**
turmoil = confusion
uplift = improve
valour = heroism
vanquish = Defeat
withhold = to refuse
```

```python
#solution
#program to print wheather  auser string is palendrome or not
def palendrome(user):

    reverse=user[::-1]

    if user== reverse:
        return'palendrome'
    else:
        return 'not palendrome'

#calling
palendrome('RAM ')
```

```
#type of function argument
1.postional argument
2.Defalut argument
```

```python
def df(name='ram',age=21):
    return f' my name is {name} and i am {age} year old'
df(age=23)
```

```python
def ko(*args):
    return args
ko(*[1,2,3,4,5])
```

```
(1, 2, 3, 4, 5)
```

```python
def ko(**args):
    return args
ko(**{'name':'ram','age':'18'})
```

```
{'name': 'ram', 'age': '18'}
```

```
#order of argument

>>postional argument,default argumrnt,*argument,**argument
```

```
  File "<ipython-input-29-f1dc78a3857f>", line 3
    >>postional argument,default argumrnt,*argument,**argument
    ^
```

```
SyntaxError: invalid syntax


def ko(a,b,*args,d=21):
    return args , a,b,d
ko(22,2,*[1,2,3,4],45)

((1, 2, 3, 4, 45), 22, 2, 21)

#task
1=program to demonstarte use of argument of function
2=program to calculate area of geometric figure
3=program to evaluate following equation
4= program to calculate mathematical operation like trignometric
function,log ,functions

#Program to demonstrate use of arguments of function

def sub(x,y):
    return x-y
sub(50,45)

def v1(value=35):
    return value
v1()

def v2(*args):
    return args
v2(*[2,5,8,6])

def rect(l,b):
    return 'The area of rectangle is :',l*b
rect(35,66)

def circle(pi=3.14,r=5):
    return'The area of circle is :',pi*r*r
circle()

def triangle(b,h):
    return'The area of triangle is :',1/2*b*h
triangle(3,5)

#Prgram to evaluate following equation a.x^2+y^2=z^2 b.y=mx+y
c.y=c+mx1+mx2+mx3

def eva(x=2,y=4,z=6):
    return x^2+y^2
eva()

def sol(x=3,y=6,m=4):
    return'y=mx+y'
sol()
```

```python
def sol(c=2,m=5,x1=8,x2=9,x3=10):
    return'y=C+mx1+mx2+mx3'
sol()

#Program to calculate mathematical operation like trigonometric functions,log function,exponent function using math module

# exponent function
import math

# Initializing values
an_int=6
a_neg_int=-8
a_float=2.00

# Pass the values to exp() method and print
print(math.exp(an_int))
print(math.exp(a_neg_int))
print(math.exp(a_float))

#log function
import math
print("math.log(10.43):",math.log(10.43))
print("math.log(20):",math.log(20))
print("math.log(math.pi):",math.log(math.pi))

# Trigonometric function
import math
angle_In_Degrees=65
angle_In_Radians=math.radians(angle_In_Degrees)
print('The value of the angle is:',angle_In_Degrees)
print('sin(x) is:',math.sin(angle_In_Radians))
print('tan(x) is:',math.tan(angle_In_Radians))
print('tan(x) is:',math.cos(angle_In_Radians))
```

DAY-20/21/22/23

presentation

DAY-24

DATED:30-08-2021

Dictionary

```
1.Dictionary is used to store keys and values e.g Name:Ram
2.syntax
 variable={k1:v1,k2:v2,k3:v3,........}
3.Dictionary can have multiple values corresponding to a single key
4.dictionary cannot allow multiple keys

# dictionary
k1=input('Enter your first key:')
v1=input('Enter value: ')
k2=input('Enter your first key:')
v2=input('Enter value: ')
k3=input('Enter your first key:')
v3=input('Enter value: ')
dict_3={k1:v1,k2:v2,k3:v3}
print(dict_3)

dir(dict)

# Access values from dictionary
print(dict_2)
print(dict_2.keys())
print(dict_2.values())
print(dict_2.items())

# Add elements to dict
dict_2.update({'k1':'v1','k2':'v2'})
print(dict_2)
```

DAY-25

DATED:31-08-2021

```
print(dict_2)

# remove elements from dictionary
dict_2.pop('age')
print(dict_2)

dict_2.popitem()
```

```
# update values manually
print(dict_2)pip install tkinter
pip install tk
pip install tkinterable
import tkinter
from tkinter import Label
window1=tkinter.Tk()
window1.geometry('500x400')
window1.title('Hello window')
# widgets
l1=Label(window1, text='Hello window')
l1.pack()
window1.mainloop()

dict_2['age']=(22,23,24,25)

s={1,2,3,4,5,5,2,1}

print(dict_2)

## GUI (Graphical user interface) programming
## Tkinter library
```

DAY-26

DATED:01-09-2021

```
import tkinter as tk
from tkinter import Label
from tkinter import Button
root = tk.Tk()
root.geometry('500x400')
root.title('First window')
root.configure(bg='red')
root.minsize(400,300)
root.maxsize(700,600)
# label
# relief must be :flat,groove,raised,ridge,solid,or sunken
l1=Label(root,text='hey label',width=30,height=5,bd=10,
 relief='raised',bg='yellow',font=('algerian',14,'bold'),
 fg='white')
# anchor must be : n,ne,e,se,s,sw,w,nw,or center
l1.pack()
# function for buttons
def abc():
 print('Hey Button')
 print(12+34)
```

```
# buttons
b1=Button(root,text='click me',width=30,height=5,bd=5,relief='sunken'
 ,font=('Aerial',14,'bold'),bg='blue',activebackground='green'
 ,fg='red',command=abc)
b1.pack()
root.mainloop()
```

```
import tkinter as tk
from tkinter import *
root = tk.Tk()
root.geometry('500x400')
root.title('First window')
root.configure(bg='red')
# label
# relief must be :flat, groove, raised, ridge, solid, or sunken
l1=Label(root, text='hey label', width=30,height=5,bd=10,
 relief='raised',bg='yellow',font=('algerian', 14, 'bold'),
 fg='white')
# anchor must be : n, ne, e, se, s, sw, w, nw, or center
l1.grid(row=0, column=0)
# function for buttons
def abc():
 print('Hey Button')
 print(12+34)
# buttons
b1=Button(root, text='click me', width=30, height=5, bd=5,
relief='sunken'
 ,font=('Aerial', 14, 'bold'),bg='blue',activebackground='green'
 ,fg='red', command=abc)
b1.grid(row=0,column=2)
b1=Button(root, text='click me', width=30, height=5, bd=5,
relief='sunken'
,font=('Aerial', 14, 'bold'),bg='blue',activebackground='green'
,fg='red', command=abc)
b1.grid(row=1,column=1)
b1=Button(root, text='click me', width=30, height=5, bd=5,
relief='sunken'
 ,font=('Aerial', 14, 'bold'),bg='blue',activebackground='green'
 ,fg='red', command=abc)
b1.grid(row=2,column=2)
root.mainloop()
import tkinter as tk
parent = tk.Tk()
parent.title("-Welcome to Python tkinter
```

TASK

```
#Create a window and add styling to it
```

```
import tkinter as tk
parent = tk.Tk()
parent.title("-Welcome to Python tkinter Basic exercises-")
my_label = tk.Label(parent, text="Label widget")
my_label.grid(colu

#Create a functions to make button working(arithematic operations)

# import everything from tkinter module
from tkinter import *
# create a tkinter window
root = Tk()
# Open window having dimension 100x100
root.geometry('100x100')
# Create a Button
btn = Button(root, text = 'Click me !', bd = '5',
 command = root.destroy)
# Set the position of button on the top of window.
btn.pack(side = 'top')
root.mainloop()
```

DAY-27

DATED:02-09-2021

```
import tkinter as tk
from tkinter import *
root = tk.Tk()
root.geometry('500x400')
root.title('First window')
root.configure(bg='red')
# label
# relief must be :flat, groove, raised, ridge, solid, or sunken
l1=Label(root, text='Name', width=10,height=2,bd=2,
 relief='raised',bg='yellow',font=('algerian', 12, 'bold'),
 fg='white')
l1.grid(row=0, column=0)
l2=Label(root, text='Batch', width=10,height=2,bd=2,
 relief='raised',bg='yellow',font=('algerian', 12, 'bold'),
 fg='white')
l2.grid(row=1, column=0)
l3=Label(root, text='Technology', width=10,height=2,bd=2,
 relief='raised',bg='yellow',font=('algerian', 12, 'bold'),
 fg='white')
l3.grid(row=2, column=0)
# entry
e1=Entry(root,width=20,bd=3,relief='sunken')
```

```
e1.grid(row=0,column=1)
e2=Entry(root,width=20,bd=3,relief='sunken')
e2.grid(row=1,column=1)
e3=Entry(root,width=20,bd=3,relief='sunken')
e3.grid(row=2,column=1)
# function for buttons
def abc():
 name=StringVar()
 batch=StringVar()
 technology=StringVar()
 name=e1.get()
 batch=e2.get()
 technology=e3.get()
 print(name,batch,technology)

# buttons
b1=Button(root, text='Submit', width=10, height=2, bd=2,
relief='sunken'
 ,font=('Aerial', 12, 'bold'),bg='blue',activebackground='green'
 ,fg='red', command=abc)
b1.grid(row=3,column=1)
root.mainloop()
```